

S2Q26

Question 26 – Stage 2

The integers 1 to 10 are shared between two people.

They each work out the product of their 5 integers.

The highest common factor of the two products is h .

What is the sum of all the different possible values of h ?

Working

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In [1]: from itertools import combinations
from math import prod, gcd

digits = set(range(1,11))
Hs = set()

for person1 in combinations(digits,r=5):
    person2 = digits.difference(person1)

    product1 = prod(person1)
    product2 = prod(person2)

    Hs.add(gcd(product1,product2))

print(f"Total: {sum(Hs)}")
```

Total: 2355